

CHAPTER TWO

OVERVIEW OF JAVA LANGUAGE

Outline

- What is Java?
- Why Java?
- A simple Java Program
- Java Virtual Machine
- **Java Variables**
- **Control Statements**
- **Loops in Java**
- Operators

WHAT IS JAVA?

- ❖ Java is
 - ❖ a high level,
 - ❖ robust, secured and
 - ❖ object-oriented programming language.
 - ❖ Java was developed by a **Green team** led by James Gosling at *SUN Microsystems*.

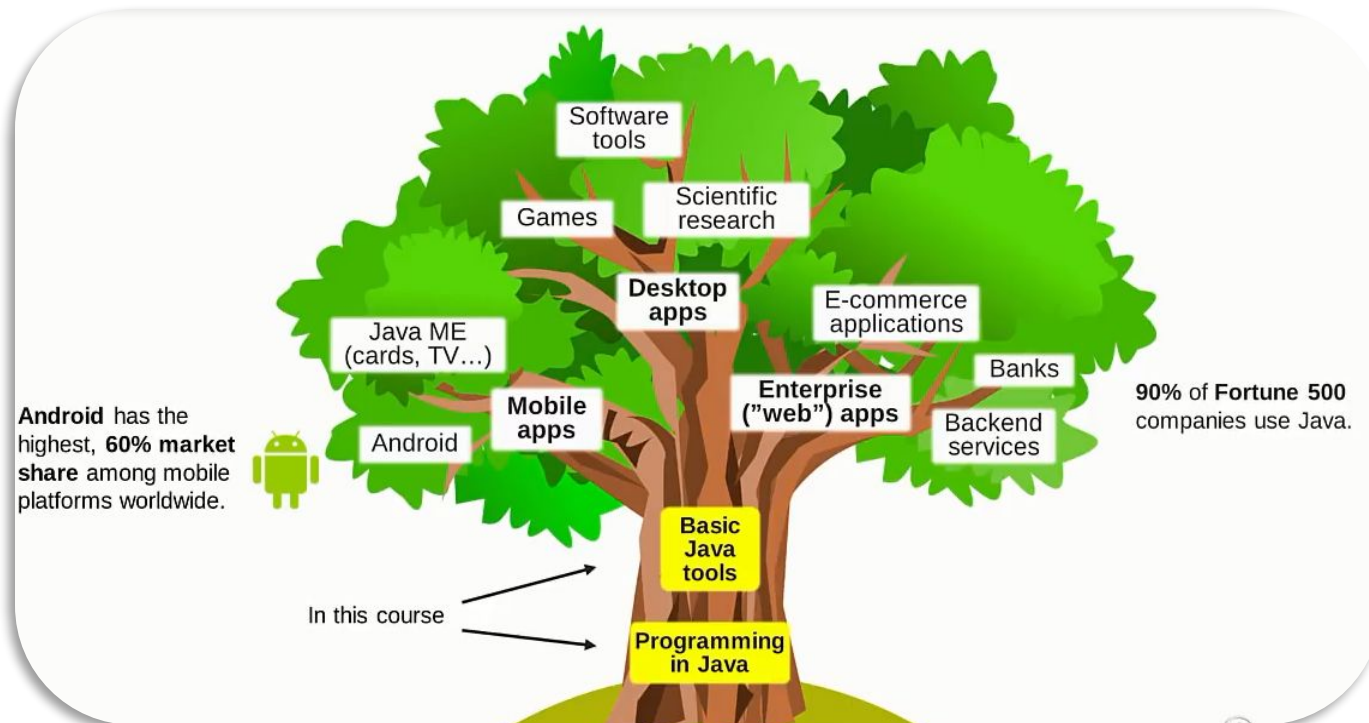
Java is popular

- In the **TOP 3** (if not the most popular)
- **3 billion** devices run...
- **Platform independency** (write once, run anywhere)
- Lots of available, very **well paying jobs**
 - **Great demand** for Java developers
- **Wide usability**



WHAT IS JAVA?.....

- ❖ The three editions of Java are:-
 - ❖ Java Micro Edition (ME) for mobile apps development.
 - ❖ Java Standard Edition (SE) for developing standalone application like desktop apps, gaming apps etc.
 - ❖ Java Enterprise Edition (EE) for large scale apps like core banking



WHY JAVA?...

- ❖ objective of **Green team/Java team** was to improve C++ compiler and make the execution of a C++ source code **platform independent**.



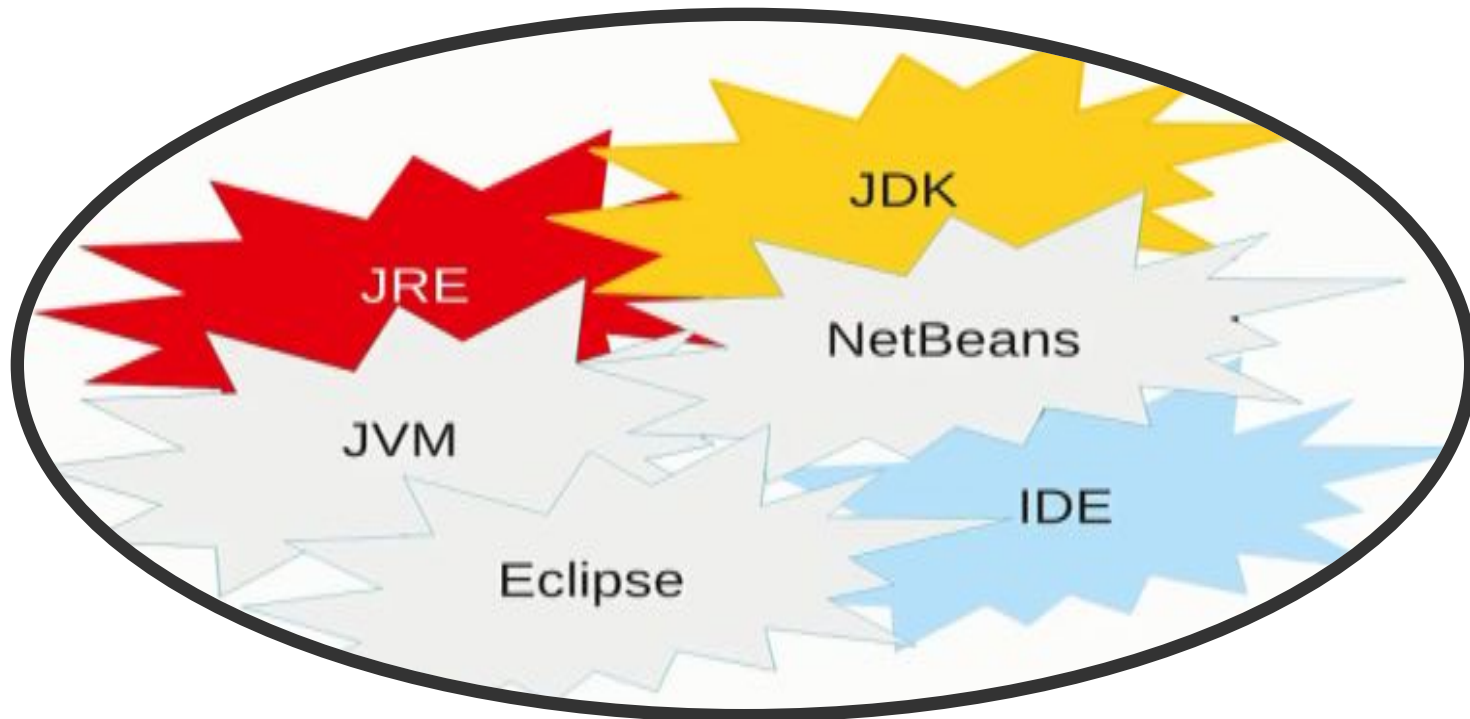
- ❖ The motto of the team was “*Write Once and Run anywhere*”
- ❖ As a result, the team has come up with the concept of **Java Virtual Machine (JVM)**

A SIMPLE JAVA PROGRAM

```
public class Welcome {  
    public static void main(String[] args){  
        System.out.println("welcome to java");  
    } }
```

JAVA VIRTUAL MACHINE

- In java, the acronyms in the figure below are very common.



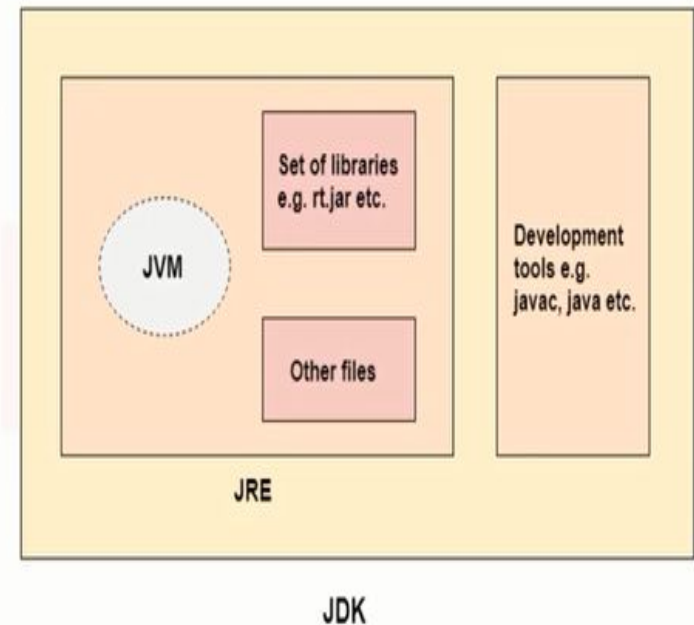
JAVA VIRTUAL MACHINE.....

JDK is an acronym for Java Development Kit. The Java Development Kit (JDK) is a software development environment which is used to develop Java applications and applets. It physically exists. It contains JRE + development tools.

JDK is an implementation of any one of the below given Java Platforms released by Oracle Corporation:

- Standard Edition Java Platform
- Enterprise Edition Java Platform
- Micro Edition Java Platform

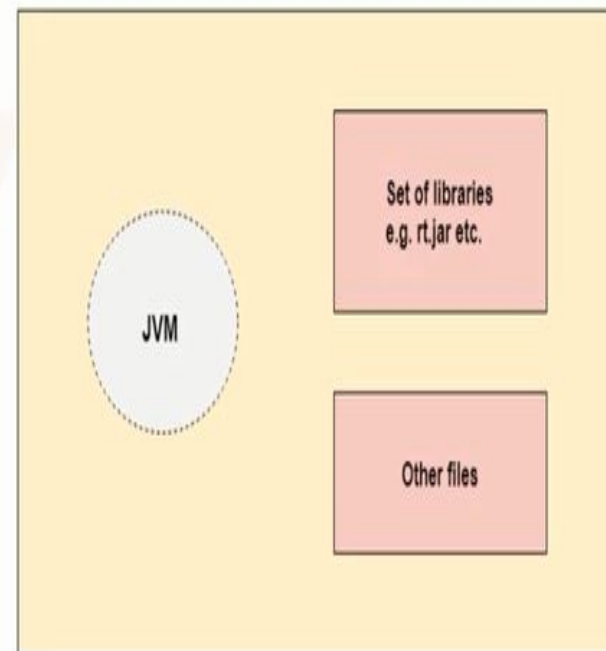
The JDK contains a private Java Virtual Machine (JVM) and a few other resources such as an interpreter/loader (java), a compiler (javac), an archiver (jar), a documentation generator (Javadoc), etc. to complete the development of a Java Application.



JAVA VIRTUAL MACHINE...

JRE is an acronym for Java Runtime Environment. It is also written as Java RTE. The Java Runtime Environment is a set of software tools which are used for developing Java applications. It is used to provide the runtime environment. It is the implementation of JVM. It physically exists. It contains a set of libraries + other files that JVM uses at runtime.

The implementation of JVM is also actively released by other companies besides Sun Micro Systems.

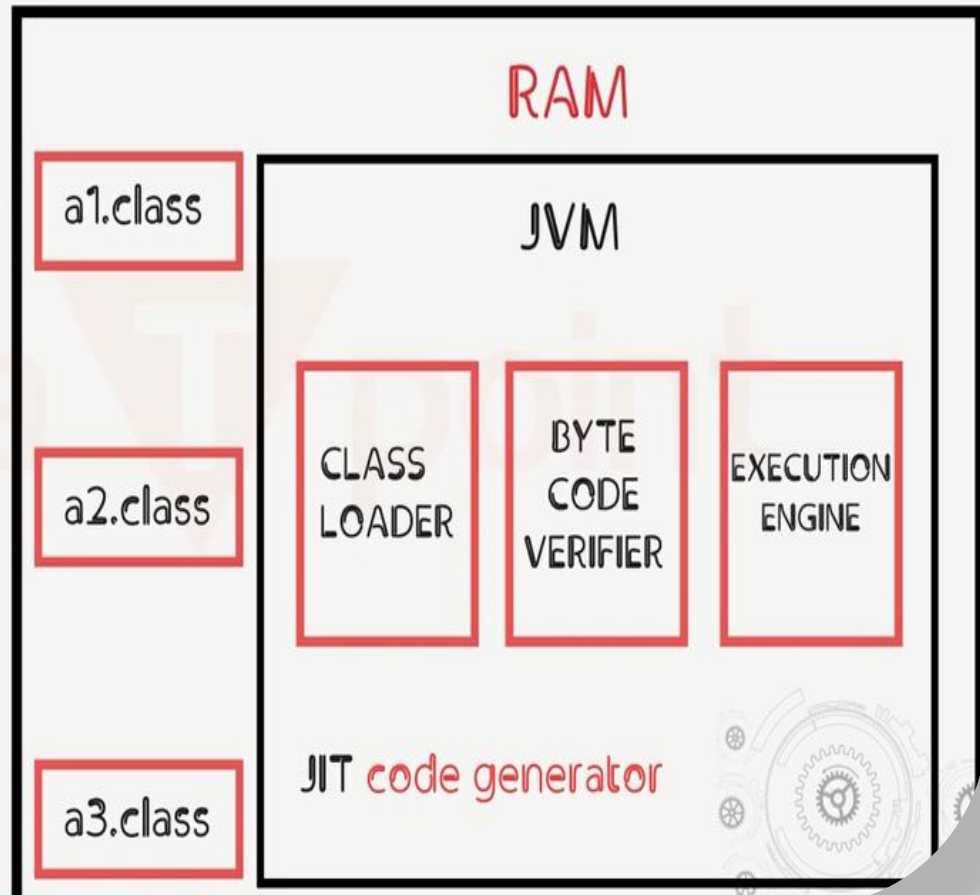


JRE

JAVA VIRTUAL MACHINE...

JAVA ARCHITECTURE

JIT converts
BYTECODE into
machine code



NetBeans

Eclipse

Java Development Kit
(JDK)

Java Runtime Environment
(JRE)

Java Virtual
Machine
(JVM)

JAVA VARIABLES

- ❖ A variable is a container which holds the value while the Java program is executed.
- ❖ There are three types of variables in java:
 1. **local,**
 2. **instance and**
 3. **static.**
- ❖ A variable is assigned with a **data type**.
- ❖ Variable is a ***name of memory location***.
- ❖ There are two types of ***data types in Java***:
 1. primitive and
 2. non-primitive.
- ❖ Java is a statically-typed programming language.
 - ❖ It means, all variable's type and name must be declared before its use.
 - ❖ Case sensitive also

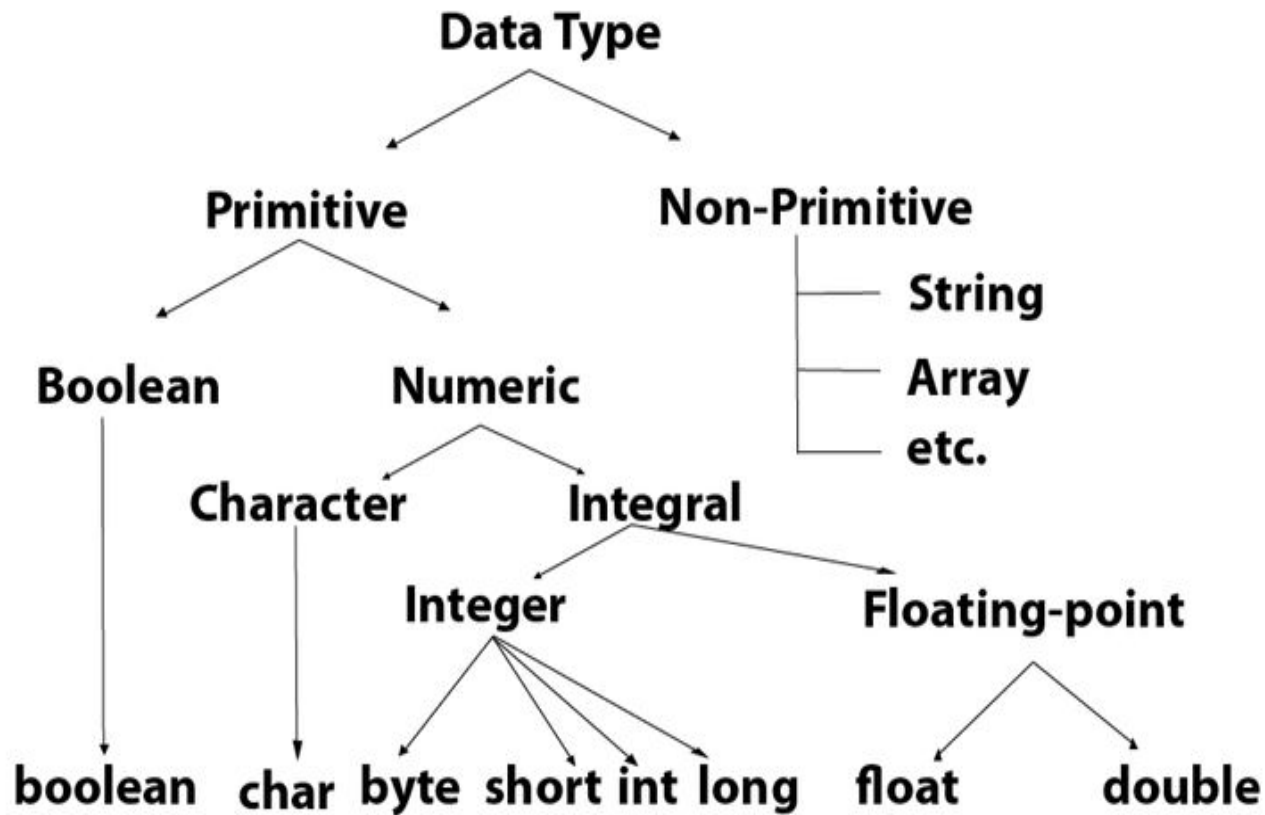
DATA TYPES IN JAVA

- ❖ Data types specify the different sizes and values that can be stored in the variable.
- ❖ There are two types of data types in Java:
 1. **Primitive data types:** The primitive data types include *boolean, char, byte, short, int, long, float and double*.
 2. **Non-primitive data types:** The non-primitive data types include Classes, Interfaces, and Arrays.

JAVA PRIMITIVE DATA TYPES

- ❖ In Java language, primitive data types are the building blocks of data manipulation.
- ❖ These are the most basic data types available in Java language.
- ❖ There are 8 types of primitive data types:
 1. boolean data type
 2. byte data type
 3. char data type
 4. short data type
 5. int data type
 6. long data type
 7. float data type
 8. double data type

DATA TYPES IN JAVA...



DATA TYPES IN JAVA...

Data Type	Default Value	Default size
boolean	false	1 bit
char	'\u0000'	2 byte
byte	0	1 byte
short	0	2 byte
int	0	4 byte
long	0L	8 byte
float	0.0f	4 byte
double	0.0d	8 byte

DATA TYPES IN JAVA...

❖ Boolean Data Type

- ❖ used to store only two mutually exclusive possible values: **true** and **false** **Example:** Boolean one = false;

❖ Byte Data Type

- ❖ It is an 8-bit signed two's complement integer.
- ❖ Its value-range lies between -128 to 127 (inclusive).
- ❖ Its minimum value is -128 and maximum value is 127.
- ❖ Its default value is 0
- ❖ The byte data type is used to save memory in large arrays where the memory savings is most required. It saves space because a byte is 4 times smaller than an integer. It can also be used in place of "int" data type.
- ❖ **Example:** byte a = 10, byte b = -20

- ❖ **Example:** short s = 10000, short r = -5000
- ❖ **Example:** int a = 100000, int b = -200000
- ❖ **Example:** float f1 = 234.5f
- ❖ **Example:** double d1 = 12.3
- ❖ **Example:** char letterA = 'A'

OPERATORS IN JAVA

- ❖ Java provides a rich set of operators to manipulate variables.
- ❖ We can divide all the Java operators into the following groups
 - ❖ Arithmetic Operators
 - ❖ Relational Operators
 - ❖ Bitwise Operators
 - ❖ Logical Operators
 - ❖ Assignment Operators
 - ❖ Miscellaneous Operators

JAVA OPERATOR PRECEDENCE

Operator Type	Category	Precedence
Unary	postfix	<code>expr++ expr--</code>
	prefix	<code>++expr --expr +expr -expr ~ !</code>
Arithmetic	multiplicative	<code>* / %</code>
	additive	<code>+ -</code>
Shift	shift	<code><< >> >>></code>
Relational	comparison	<code>< > <= >= instanceof</code>
	equality	<code>== !=</code>
Bitwise	bitwise AND	<code>&</code>
	bitwise exclusive OR	<code>^</code>
	bitwise inclusive OR	<code> </code>
Logical	logical AND	<code>&&</code>
	logical OR	<code> </code>
Ternary	ternary	<code>? :</code>
Assignment	assignment	<code>= += -= *= /= %= &= ^= = <<= >>= >>>=</code>

CONTROL STATEMENTS

❖ Java If-else Statement

- ❖ The Java *if statement* is used to test the condition.
- ❖ It checks boolean condition: *true* or *false*.
- ❖ There are various types of *if statement* in Java.
 - ❖ if statement
 - ❖ if-else statement
 - ❖ if-else-if ladder
 - ❖ nested if statement

JAVA SWITCH STATEMENT

- ❖ The Java *switch statement* executes one statement from multiple conditions.
- ❖ It is like if-else-if ladder statement.
- ❖ The switch statement works with byte, short, int, long..
- ❖ the switch statement tests the equality of a variable against multiple values.
- ❖ **Points to Remember**
- ❖ There can be *one or N number of case values* for a switch expression.
 - ❖ The case value must be of switch expression type only.
 - ❖ The case value must be *literal or constant*.
 - ❖ It doesn't allow variables.
- ❖ The case values must be *unique*.
- ❖ In case of duplicate value, it renders compile-time error.
- ❖ The Java switch expression must be of *byte, short, int, long..*
- ❖ Each case statement can have a *break statement* which is optional. When control reaches to the break statement, it jumps the control after the switch expression.
- ❖ If a break statement is not found, it executes the next case.
- ❖ The case value can have a *default label* which is optional.

LOOPS IN JAVA

- ❖ In programming languages, loops are used to execute a set of instructions/functions repeatedly when some conditions become true.
- ❖ There are three types of loops in Java.
 1. for loop
 2. while loop
 3. do-while loop

LOOPS IN JAVA..

Java For Loop vs While Loop vs Do While Loop

Comparison	for loop	while loop	do while loop
Introduction	The Java for loop is a control flow statement that iterates a part of the programs multiple times.	The Java while loop is a control flow statement that executes a part of the programs repeatedly on the basis of given boolean condition.	The Java do while loop is a control flow statement that executes a part of the programs at least once and the further execution depends upon the given boolean condition.
When to use	If the number of iteration is fixed, it is recommended to use for loop.	If the number of iteration is not fixed, it is recommended to use while loop.	If the number of iteration is not fixed and you must have to execute the loop at least once, it is recommended to use the do-while loop.
Syntax	<pre>for (init; condition; incr/decr) { // code to be executed }</pre>	<pre>while (condition) { //code to be executed }</pre>	<pre>do { //code to be executed } while (condition);</pre>
Example	<pre>//for loop for (int i=1; i<=10; i++) { System.out.println(i); }</pre>	<pre>//while loop int i=1; while (i<=10) { System.out.println(i); i++; }</pre>	<pre>//do-while loop int i=1; do { System.out.println(i); i++; } while (i<=10);</pre>

LOOPS IN JAVA..

Syntax for infinite loop	<pre>for(;;){ //code to be executed }</pre>	<pre>while(true){ //code to be executed }</pre>	<pre>do{ //code to be executed }while(true);</pre>
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LOOPS IN JAVA...

❖ **Java Nested For Loop**

- ❖ If we have a for loop inside the another loop, it is known as nested for loop.
- ❖ The inner loop executes completely whenever outer loop executes a single iteration.
- ❖ The for-each loop is used to traverse array or collection in java.
- ❖ It is easier to use than simple for loop because we don't need to increment value and use subscript notation.
- ❖ It works on elements basis not index.
- ❖ It returns element one by one in the defined variable.
 - ❖ **Syntax:**
 - ❖ `for(Type var:array){`
 - ❖ `//code to be executed`
 - ❖ `}`

JAVA ARRAYS

- ❖ Normally, an array is a collection of similar type of elements which has contiguous memory location.
- ❖ **Java array** is an object which contains elements of a similar data type.
- ❖ Additionally,
- ❖ The elements of an array are stored in a contiguous memory location.
- ❖ It is a data structure where we store similar elements.
- ❖ We can store only a fixed set of elements in a Java array.
- ❖ Array in Java is index-based, the first element of the array is stored at the 0th index, 2nd element is stored on 1st index and so on.

JAVA ARRAYS...

- ❖ In Java, array is an object of a dynamically generated class
- ❖ **Types of Array in java**
 - ❖ There are two types of array.
 1. Single Dimensional Array
 2. Multidimensional Array
- ❖ **Single Dimensional Array in Java**
 - ❖ **Syntax to Declare an Array in Java**
 - ❖ `dataType[] arr; (or)`
 - ❖ `dataType []arr; (or)`
 - ❖ `dataType arr[];`

JAVA ARRAYS...

- ❖ **Declaration, Instantiation and Initialization of Java Array**
- ❖ We can declare, instantiate and initialize the java array together by:
- ❖ `int a[]={33,3,4,5};`//declaration, instantiation and initialization

JAVA ARRAYS...

❖ **Multidimensional Array in Java**

- ❖ In such case, data is stored in row and column based index (also known as matrix form).

❖ **Syntax to Declare Multidimensional Array in Java**

- ❖ `dataType[][] arrayRefVar; (or)`
- ❖ `dataType [][]arrayRefVar; (or)`
- ❖ `dataType arrayRefVar[][]; (or)`
- ❖ `dataType []arrayRefVar[];`

❖ **Example to instantiate Multidimensional Array in Java**

- ❖ `int[][] arr=new int[3][3];//3 row and 3 column`

```
Command Prompt
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C:\Users\GUTAM>cd Desktop

C:\Users\GUTAM\Desktop>cd Test

C:\Users\GUTAM\Desktop\Test>dir
Volume in drive C has no label.
Volume Serial Number is B803-7CEB

Directory of C:\Users\GUTAM\Desktop\Test

14-Dec-20  10:27 AM    <DIR>          .
14-Dec-20  10:27 AM    <DIR>          ..
14-Dec-20  10:36 AM                425 WelcomeT.class
14-Dec-20  10:36 AM                216 WelcomeT.java
                2 File(s)                641 bytes
                2 Dir(s)  22,255,034,368 bytes free

C:\Users\GUTAM\Desktop\Test>java WelcomeT
welcome to java

C:\Users\GUTAM\Desktop\Test>cd\

C:\>_
```